

(12) **United States Patent**
Huang et al.

(10) **Patent No.:** **US 12,414,395 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **PHOTOSENSITIVE DEVICE**

(71) Applicant: **Au Optronics Corporation**, Hsinchu (TW)

(72) Inventors: **Yen-Tang Huang**, Hsinchu (TW);
Yan-Liang Chen, Hsinchu (TW)

(73) Assignee: **Au Optronics Corporation**, Hsinchu (TW)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 540 days.

(21) Appl. No.: **17/722,409**

(22) Filed: **Apr. 18, 2022**

(65) **Prior Publication Data**

US 2022/0375979 A1 Nov. 24, 2022

Related U.S. Application Data

(60) Provisional application No. 63/224,977, filed on Jul. 23, 2021, provisional application No. 63/189,960, filed on May 18, 2021.

(30) **Foreign Application Priority Data**

Dec. 2, 2021 (TW) 110144977

(51) **Int. Cl.**

H10F 39/00 (2025.01)

G02B 27/30 (2006.01)

G06V 40/13 (2022.01)

(52) **U.S. Cl.**

CPC **H10F 39/8063** (2025.01); **G02B 27/30** (2013.01); **H10F 39/8053** (2025.01); **G06V 40/1318** (2022.01)

(58) **Field of Classification Search**

CPC H01L 27/14621; H01L 27/14627; G02B 27/30; G02B 3/0062; G06V 40/1318; H10F 39/8053; H10F 39/8063

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2005/0061950 A1* 3/2005 Jiang H10F 39/8063 250/214.1

2005/0274968 A1* 12/2005 Kuo H10F 39/8053 257/98

(Continued)

FOREIGN PATENT DOCUMENTS

CN 108681703 10/2018

CN 112882279 6/2021

JP 2007047569 2/2007

Primary Examiner — Meiya Li

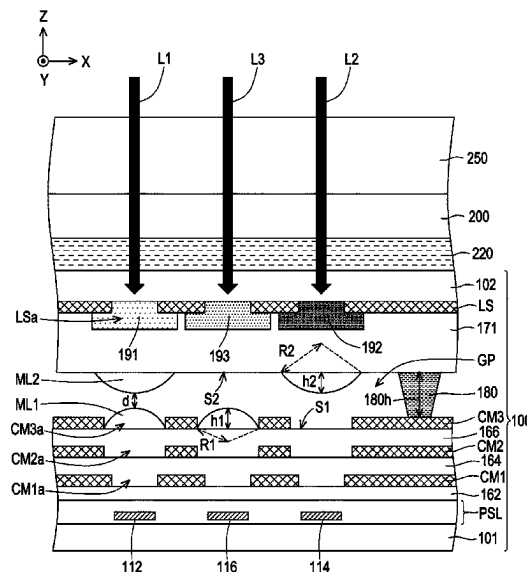
(74) *Attorney, Agent, or Firm* — JCIPRNET

(57)

ABSTRACT

A photosensitive device includes a first substrate, a second substrate, a supporting structure, multiple first microlenses, multiple second microlenses, a first photosensitive element, a second photosensitive element, and a collimating structure. The second substrate is opposite to the first substrate, and there is a gap between the first substrate and the second substrate. The supporting structure is located in the gap between the first substrate and the second substrate. The first microlenses and the second microlenses are respectively disposed on a first side and a second side of the gap. The first photosensitive element is overlapping with one of the first microlenses and one of the second microlenses. The second photosensitive element is overlapping with another one of the second microlenses. The collimating structure is located between the first substrate and the first microlenses.

17 Claims, 6 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

2012/0170072	A1	7/2012	Miyazaki et al.	
2019/0278152	A1 *	9/2019	Wang	G02F 1/133526
2020/0203532	A1 *	6/2020	Yamazaki	G02F 1/136
2021/0286963	A1	9/2021	Gao et al.	

* cited by examiner

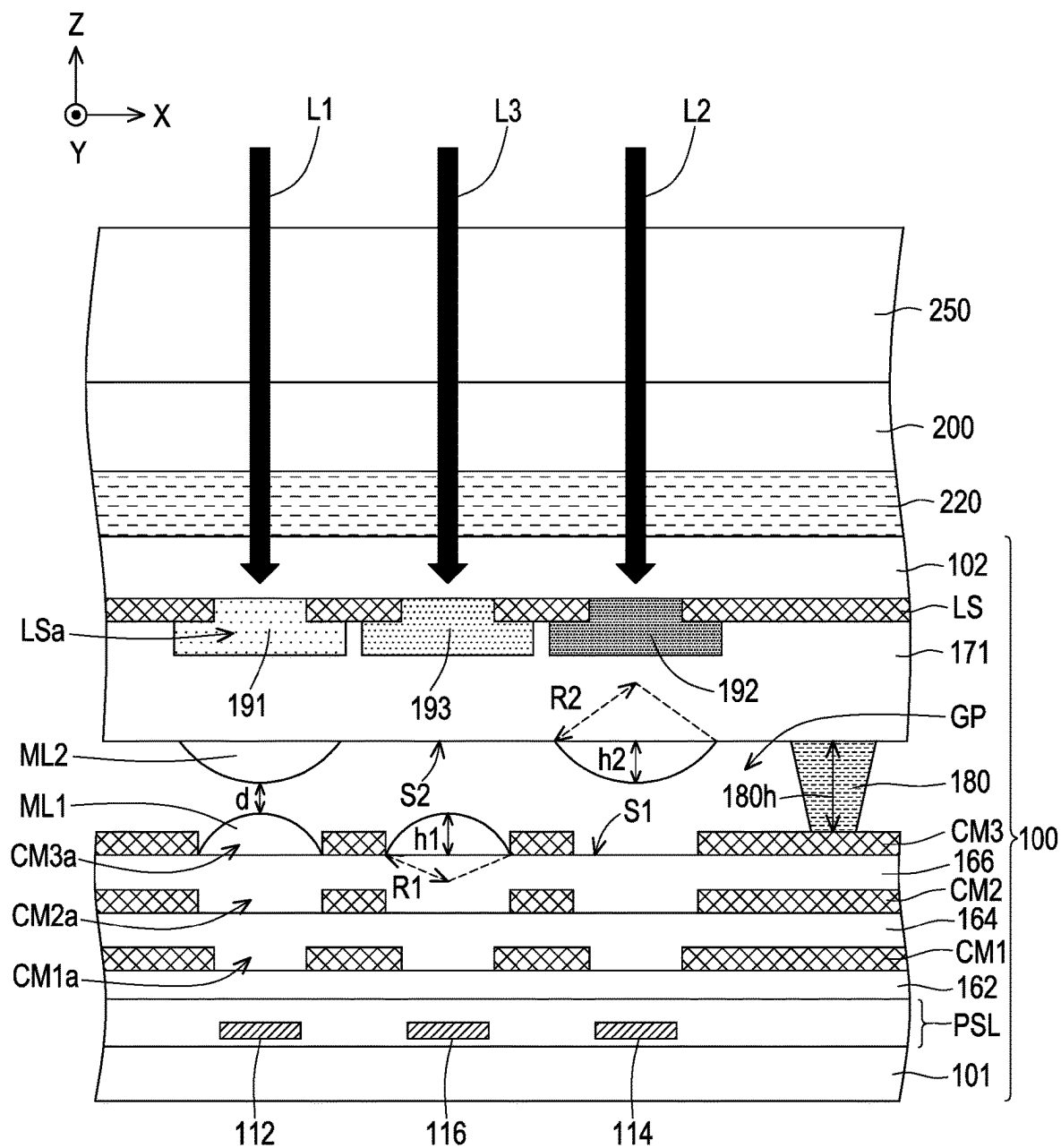


FIG. 1

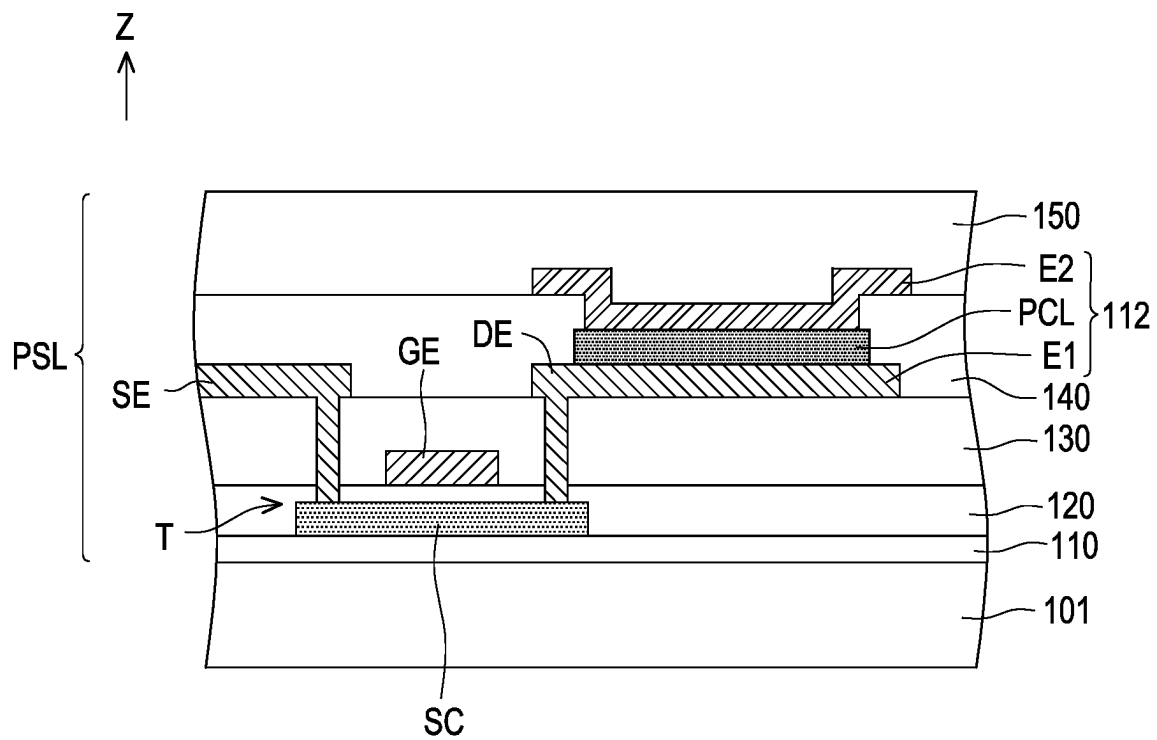


FIG. 2

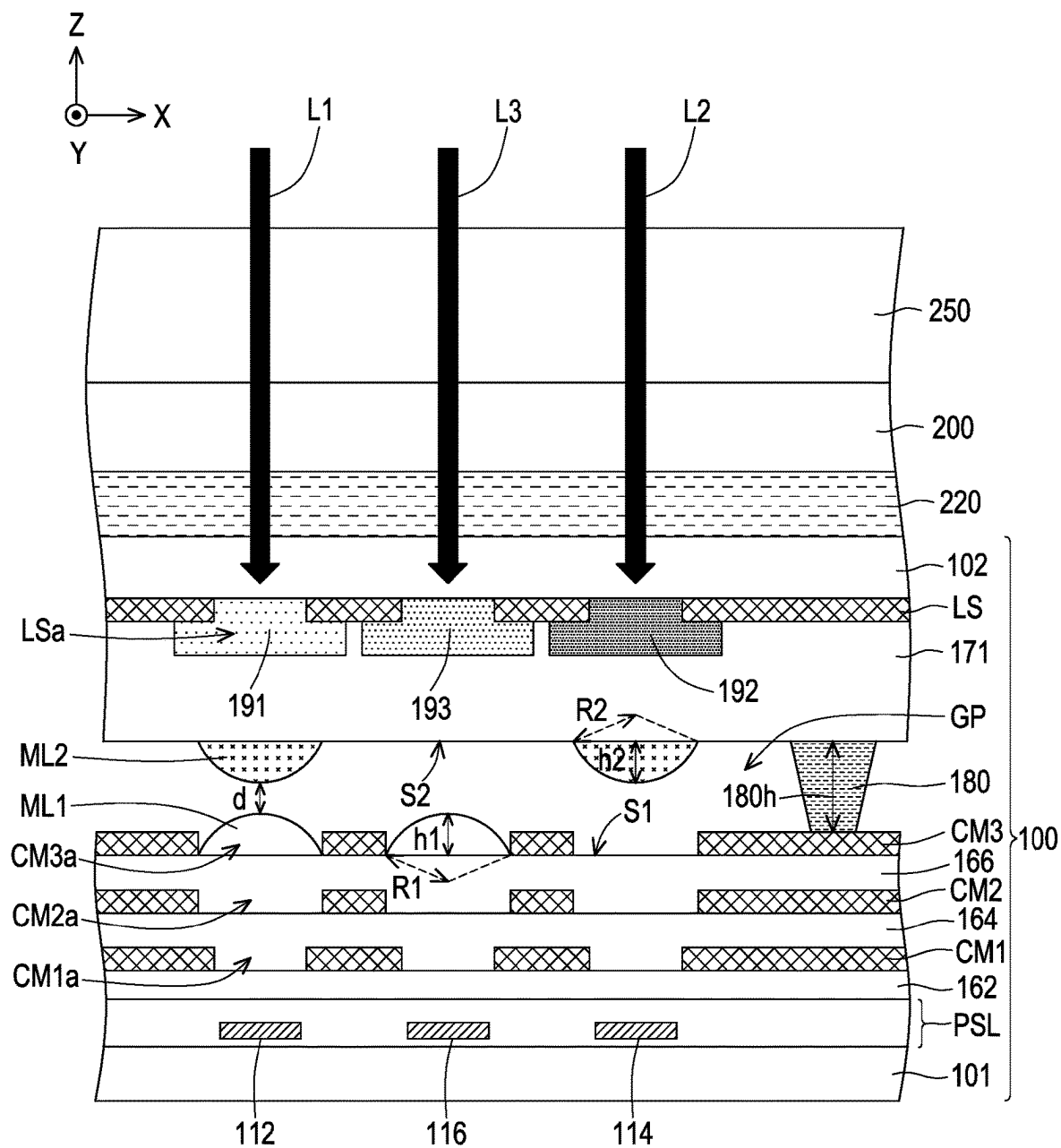


FIG. 3



FIG. 4

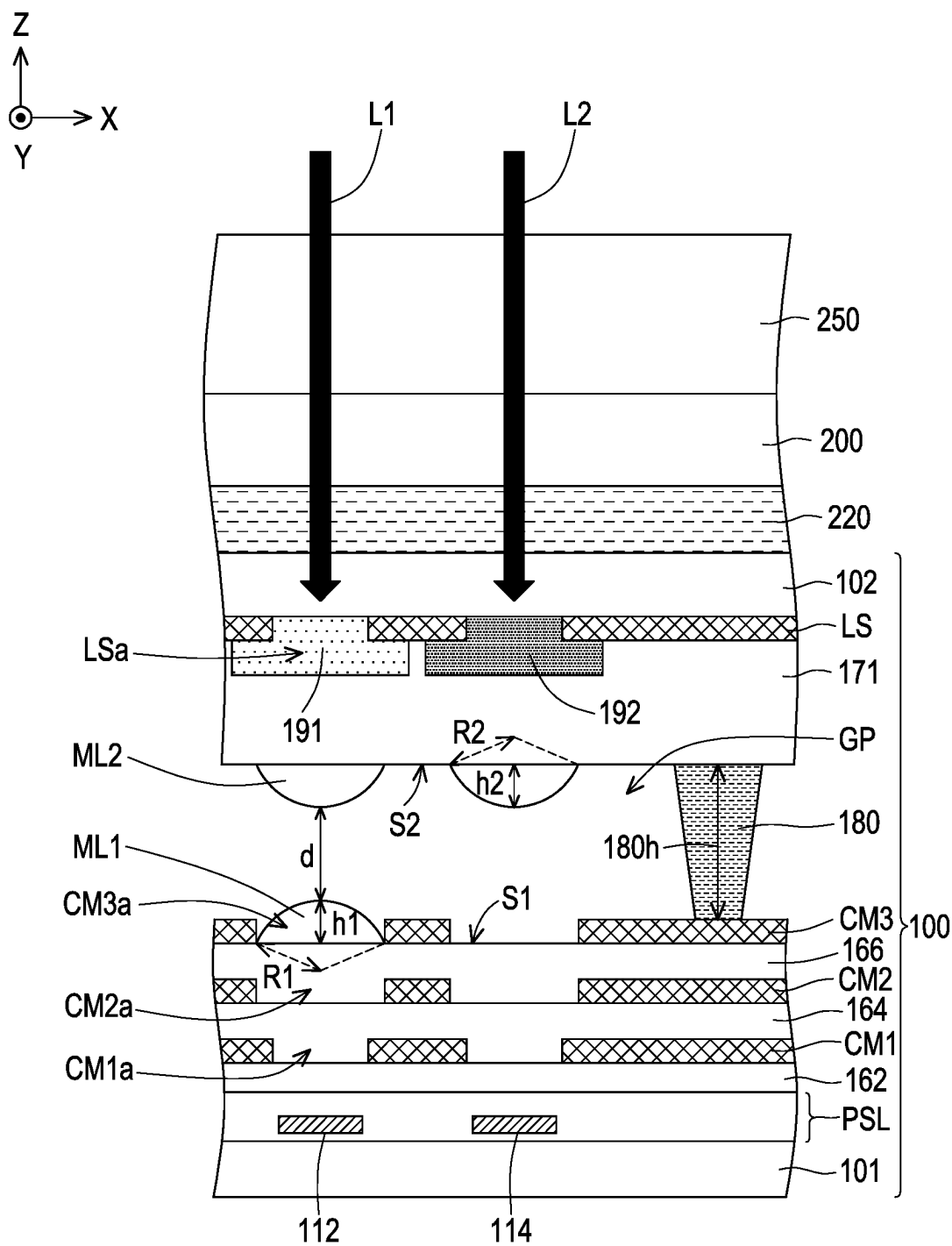


FIG. 5

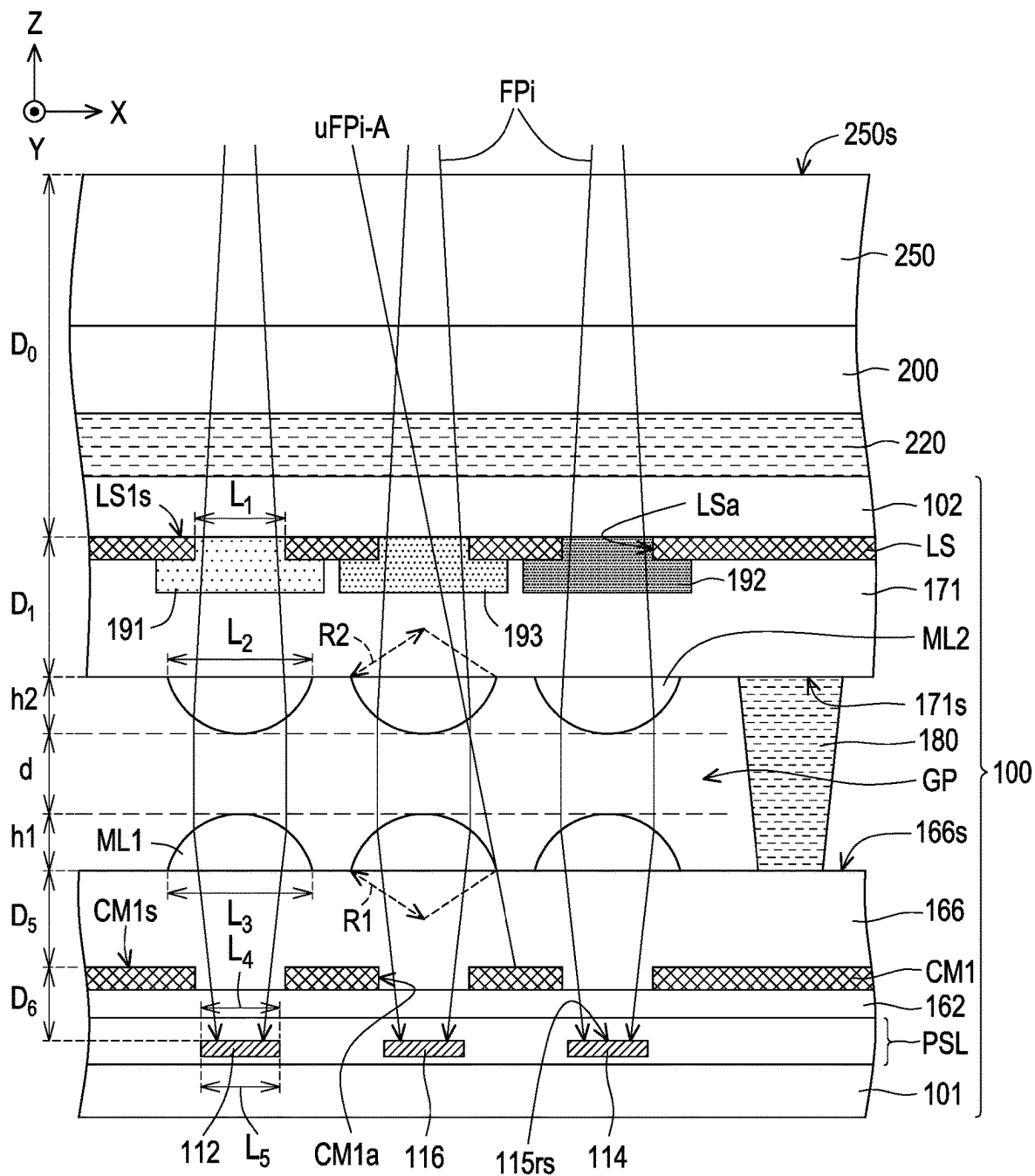


FIG. 6

1

PHOTOSENSITIVE DEVICE**CROSS-REFERENCE TO RELATED APPLICATION**

This application claims the priority benefit of U.S. provisional application Ser. No. 63/189,960, filed on May 18, 2021, U.S. provisional application Ser. No. 63/224,977, filed on Jul. 23, 2021, and Taiwan application serial no. 110144977, filed on Dec. 2, 2021. The entirety of each of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND**Technical Field**

The disclosure relates to a photosensitive device, and more particularly, to a photosensitive device having a microlens.

Description of Related Art

In order to increase a screen-to-body ratio of a display to achieve a narrow border design, the fingerprint on display sensing technology has become a trend. To put it simply, the fingerprint on display sensing technology is to dispose a photosensitive device under a display panel of an electronic device. After the electronic device detects that a user touches a display screen, the electronic device controls the display panel to emit light to illuminate a surface of a finger of the user. The sensing light is reflected (diffused) by the finger of the user into the photosensitive device under the display panel, and the reflected light is converged on a photosensitive element through multiple microlenses and a collimating structure to be converted into a digital image signal, which may obtain a fingerprint image of the user.

Generally speaking, if a color image is to be recognized, the photosensitive element for receiving the light having different wavelengths is usually required to be disposed in the photosensitive device. However, in the same material, the light having different wavelengths has different refractive indices. Specifically, the longer the wavelength of the light, the less the refractive index in a medium. As a result, the light of different colors has different focus depths in the photosensitive device, and the image recognition and light collecting ability of the photosensitive device is deteriorated.

SUMMARY

The disclosure provides a photosensitive device with good image sensing quality.

At least one embodiment of the disclosure provides a photosensitive device, including a first substrate, a second substrate, a supporting structure, multiple first microlenses, multiple second microlenses, a first photosensitive element, a second photosensitive element, and a collimating structure. The second substrate is opposite to the first substrate, and there is a gap between the first substrate and the second substrate. The support supporting is located in the gap between the first substrate and the second substrate. The first microlenses and the second microlenses are respectively disposed on a first side and a second side of the gap. The first photosensitive element is overlapping with one of the first microlenses and one of the second microlenses. The second photosensitive element is overlapping with another one of

2

the second microlenses. The collimating structure is located between the first substrate and the first microlens.

At least one embodiment of the disclosure provides a photosensitive device including a first substrate, a second substrate, multiple first microlenses, multiple second microlenses, a first photosensitive element, and a second photosensitive element. The second substrate is opposite to the first substrate, and there is a gap between the first substrate and the second substrate. The first microlenses and the second microlenses are respectively disposed on a first side and a second side of the gap. A material of the first microlenses is different from a material of the second microlenses, and/or a radius of curvature of the first microlenses is different from a radius of curvature of the second microlenses. The first photosensitive element is overlapping with one of the first microlenses. The second photosensitive element is overlapping with one of the second microlenses, and is not overlapping with the first microlenses.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic cross-sectional view of a display device according to an embodiment of the disclosure.

FIG. 2 is a schematic partial enlarged view of a photosensitive device of FIG. 1.

FIG. 3 is a schematic cross-sectional view of a display device according to an embodiment of the disclosure.

FIG. 4 is a schematic cross-sectional view of a display device according to an embodiment of the disclosure.

FIG. 5 is a schematic cross-sectional view of a display device according to an embodiment of the disclosure.

FIG. 6 is a schematic cross-sectional view of a display device according to an embodiment of the disclosure.

DETAILED DESCRIPTION OF DISCLOSED EMBODIMENTS

The term “about”, “approximately”, “essentially” or “substantially” used herein includes the value and an average value within an acceptable deviation range of specific values determined by a person of ordinary skill in the art, taking into account discussed measurements and a specific number of measurement-related errors (i.e., limitations of a measuring system). For example, the term “about” may mean being within one or more standard deviations of the value, or within, for example, $\pm 30\%$, $\pm 20\%$, $\pm 10\%$, and $\pm 5\%$. Moreover, the term “about”, “approximately”, “essentially” or “substantially” used herein may mean selecting a more acceptable deviation range or standard deviations according to measurement properties, cutting properties or other properties, without applying a single standard deviation to all properties.

In the accompanying drawings, thicknesses of layers, films, panels, regions, etc., are enlarged for clarity. It should be understood that when an element such as the layer, film, region, or substrate is described as being “on” or “connected to” another element, it may be directly on or connected to another element, or there may be an intervening element therebetween. In contrast, when an element is described as being “directly on” or “directly connected to” another element, there are no intervening elements therebetween. As used herein, “connected” may refer to a physical and/or electrical connection. Furthermore, “electrically connected” may refer to the existence of other elements between the two elements.

Reference will now be made in detail to the exemplary embodiments of the disclosure, and examples of the exem-

3

plary embodiments are illustrated in the accompanying drawings. Whenever possible, the same reference numerals are used in the drawings and descriptions to indicate the same or similar parts.

FIG. 1 is a schematic cross-sectional view of a display device according to an embodiment of the disclosure. FIG. 2 is a schematic partial enlarged view of a photosensitive device of FIG. 1.

Referring to FIGS. 1 and 2, a display device 10 includes a photosensitive device 100, a display panel 200, and a cover plate 250 that are disposed on top of one another. For example, the photosensitive device 100 and the display panel 200 are bonded together by an optical adhesive layer 220 distributed over an entire surface. A material of the optical adhesive layer 220 includes, for example, an optical clear resin (OCR), an optical clear adhesive (OCA), a pressure sensitive adhesive (PSA), or other suitable adhesive materials.

In this embodiment, the photosensitive device 100 may be disposed on a back side of the display panel 200. For example, the photosensitive device 100 is a fingerprint recognition module, and the display device 10 may be a fingerprint on display recognition device. However, the disclosure is not limited thereto.

The display panel 200 is, for example, an organic light emitting diode (OLED) panel, a micro light emitting diode (micro-LED) panel, a mini light emitting diode (mini-LED) panel, or other suitable self-luminous display panels. It is particularly noted that in this embodiment, the display panel 200 may also be used as an illumination light source for fingerprint recognition. However, the disclosure is not limited thereto. In other embodiments, the display panel may also be a non-self-luminous display panel (e.g., a liquid crystal display panel), and the display device uses a back-light source to provide illumination light required for the fingerprint recognition.

The photosensitive device 100 includes a first substrate 101, a second substrate 102, a supporting structure 180, multiple first microlenses ML1, multiple second microlenses ML2, a photosensitive element layer PSL, and a first collimating structure CM1. In this embodiment, the photosensitive device 100 further includes a second collimating structure CM2, a third collimating structure CM3, a light shielding pattern layer LS, a first filter element 191, a second filter element 192, and a third filter element 193.

The first substrate 101 includes a transparent or opaque substrate. For example, the first substrate 101 includes glass, quartz, organic polymers, opaque/reflective materials (e.g., conductive materials, metals, wafers, ceramics, or other applicable materials), or other applicable materials. If the conductive materials or metals are used, an insulating layer (not shown) is covered on the first substrate 101 to avoid an issue of short circuit.

The photosensitive element layer PSL is located above the first substrate 101 and includes multiple active elements and multiple photosensitive elements. In FIG. 1, a first photosensitive element 112, a second photosensitive element 114, and a third photosensitive element 116 in the photosensitive element layer PSL are shown, and the active elements electrically connected to the first photosensitive element 112, the second photosensitive element 114, and the third photosensitive element 116 are omitted. In some embodiments, the first photosensitive element 112, the second photosensitive element 114, and the third photosensitive element 116 are formed together, and the first photosensitive element 112, the second photosensitive element 114, and the third photosensitive element 116 all have the same vertical

4

distance from the first substrate 101. In other words, in some embodiments, based on a surface of the first substrate 101, the first photosensitive element 112, the second photosensitive element 114, and the third photosensitive element 116 are located at the same level height.

The first photosensitive element 112, the second photosensitive element 114, and the third photosensitive element 116 are respectively electrically connected to the corresponding active elements. In order to illustrate the electrical connection between the photosensitive elements and the active elements, in FIG. 2, the first photosensitive element 112 is taken as an example for illustration, and the second photosensitive element 114 and the third photosensitive element 116 are also electrically connected to the corresponding active elements in the same manner as in FIG. 2.

Referring to FIG. 2, the first photosensitive element 112 (which may also be the second photosensitive element 114 or the third photosensitive element 116) is electrically connected to an active element T. The active element T has a source SE, a drain DE, a gate GE, and a semiconductor pattern SC. In this embodiment, the semiconductor pattern SC is a single-layer or multi-layer structure, which includes amorphous silicon, polycrystalline silicon, microcrystalline silicon, single crystal silicon, organic semiconductor materials, oxide semiconductor materials (e.g., indium zinc oxide, indium gallium zinc oxide, other suitable materials, or a combination of the above), other suitable materials, or a combination of the above, but the disclosure is not limited thereto. The semiconductor pattern SC is located on the first substrate 101. In this embodiment, a buffer layer 110 is selectively disposed between the semiconductor pattern SC and the first substrate 101. The gate GE is overlapping with the semiconductor pattern SC, and a gate insulating layer 120 is sandwiched between the gate GE and the semiconductor pattern SC. In this embodiment, the gate GE may be selectively disposed above the semiconductor pattern SC to form a top-gate thin film transistor, but the disclosure is not limited thereto. In other embodiments, the gate GE may also be disposed under the semiconductor pattern SC to form a bottom-gate thin film transistor. An interlayered insulating layer 130 is located on the gate GE and the gate insulating layer 120. The source SE and the drain DE are located on the interlayered insulating layer 130, and are electrically connected to the semiconductor pattern SC.

A planar layer 140 is located on the interlayered insulating layer 130. The first photosensitive element 112 (which may also be the second photosensitive element 114 or the third photosensitive element 116) is located on the interlayered insulating layer 130, and is electrically connected to the drain DE of the active element T. In this embodiment, the first photosensitive element 112 (which may also be the second photosensitive element 114 or the third photosensitive element 116) includes a first electrode E1, a photoelectric conversion layer PCL, and a second electrode E2. The photoelectric conversion layer PCL has a single-layer or multi-layer structure. For example, the photoelectric conversion layer PCL is silicon-rich oxide (SRO). In other embodiments, the photoelectric conversion layer PCL is a stacked layer of a P-type semiconductor, an intrinsic semiconductor, and an N-type semiconductor. A planar layer 150 is located on the second electrode E2.

Continuing to refer to FIG. 1, the first collimating structure CM1, the second collimating structure CM2, and the third collimating structure CM3 are located above the first substrate 101. The first collimating structure CM1 has multiple first voids CM1a corresponding to the photosensitive elements (including the first photosensitive element 112,

5

the second photosensitive element **114**, and the third photosensitive element **116**). The second collimating structure **CM2** has multiple second voids **CM2a** corresponding to the photosensitive elements. The third collimating structure **CM3** has multiple third voids **CM3a** corresponding to the photosensitive elements. The first voids **CM1a**, the second voids **CM2a**, and third voids **CM3a** overlap in a direction **Z**. In this embodiment, the voids may be arranged in an array, for example, respectively arranged in multiple columns and multiple rows along a direction **X** and a direction **Y**, but the disclosure is not limited thereto.

In this embodiment, a planar layer **162** is located on the photosensitive element layer **PSL**, and the first collimating structure **CM1** is located on the planar layer **162**. A planar layer **164** is located on the first collimating structure **CM1**, and the second collimating structure **CM2** is located on the planar layer **164**. A planar layer **166** is located on the second collimating structure **CM2**, and the third collimating structure **CM3** is located on the planar layer **166**.

The first microlenses **ML1** are disposed on the first substrate **101**. The first collimating structure **CM1** and the second collimating structure **CM2** are located between the first substrate **101** and the first microlenses **ML1**.

In this embodiment, the first microlenses **ML1** may be arranged in multiple columns and multiple rows along the direction **X** and the direction **Y**. In this embodiment, some of the third voids **CM3a** are provided with the first microlenses **ML1**, while the other third voids **CM3a** are not provided with the first microlenses **ML1**. Therefore, in this embodiment, some photosensitive elements (e.g., the first photosensitive element **112** and the third photosensitive element **116**) are overlapping with the first microlenses **ML1** in the direction **Z**, while some photosensitive elements (e.g., the second photosensitive element **114**) are not overlapping with the first microlenses **ML1** in the direction **Z**.

In some embodiments, the first microlenses **ML1** may be formed by organic photoresist materials, but the disclosure is not limited thereto.

The second substrate **102** is opposite to the first substrate **101**. The second substrate **102** includes a transparent substrate. For example, the second substrate **102** includes glass, quartz, organic polymers, or other applicable materials.

The light shielding pattern layer **LS** is disposed on the second substrate **102**. For example, the light shielding pattern layer **LS** is disposed between the second substrate **102** and the first substrate **101**. The light shielding pattern layer **LS** has multiple voids **LSa**. In this embodiment, the voids **LSa** of the light shielding pattern layer **LS** are overlapping with the first voids **CM1a**, the second voids **CM2a**, and the third voids **CM3a** in the direction **Z**.

The first filter element **191**, the second filter element **192**, and the third filter element **193** are disposed on the second substrate **102**. For example, the first filter element **191**, the second filter element **192**, and the third filter element **193** are disposed between the second substrate **102** and the first substrate **101**, and are respectively filled in the corresponding voids **LSa**.

The first filter element **191** is overlapping with the first photosensitive element **112**, and is configured to enable a light **L1** having a first wavelength λ_1 to pass through. The second filter element **192** is overlapping with the second photosensitive element **114**, and is configured to enable a light **L2** having a second wavelength λ_2 to pass through. The third filter element **193** is overlapping with the third photosensitive element **116**, and is configured to enable a light **L3** having a third wavelength λ_3 to pass through.

6

In this embodiment, the first wavelength λ_1 is greater than the third wavelength λ_3 , and the third wavelength λ_3 is greater than the second wavelength λ_2 , but the disclosure is not limited thereto. In other embodiments, the first wavelength λ_1 is greater than the second wavelength λ_2 , and the second wavelength λ_2 is greater than the third wavelength λ_3 . In some embodiments, the light **L1**, the light **L2**, and the light **L3** are visible light, such as green light, blue light, and red light, but the disclosure is not limited thereto. In other embodiments, the light **L1**, the light **L2**, and the light **L3** may also include non-visible light, such as infrared light.

In some embodiments, the first filter element **191**, the second filter element **192**, and the third filter element **193** are disposed to convert the light into the light having the predetermined wavelength, but the disclosure is not limited thereto. In other embodiments, the first filter element **191**, the second filter element **192**, and the third filter element **193** are not required to be disposed, and the light having the predetermined wavelength may be obtained by selecting a light source that may emit the light having the predetermined wavelength.

A cladding layer **171** is disposed on the first filter element **191**, the second filter element **192**, and the third filter element **193**.

The second microlenses **ML2** are disposed on the second substrate **102**. At least one of the first filter element **191**, the second filter element **192**, and the third filter element **193** is located between the second substrate **102** and the second microlenses **ML2**.

In this embodiment, the second microlenses **ML2** may be disposed in multiple columns and multiple rows along the direction **X** and the direction **Y**. In this embodiment, some photosensitive elements (e.g., the second photosensitive element **114** and the first photosensitive element **112**) are overlapping with the second microlenses **ML2** in the direction **Z**, and some photosensitive elements (e.g., the third photosensitive element **116**) are not overlapping with the second microlenses **ML2** in the direction **Z**.

In some embodiments, the second microlenses **ML2** may be formed by the organic photoresist materials, but the disclosure is not limited thereto.

In this embodiment, there is a gap **GP** between the first substrate **101** and the second substrate **102**, and the gap **GP** is, for example, an air gap. The supporting structure **180** is located in the gap **GP** between the first substrate **101** and the second substrate **102**. It may also be said that a thickness of the gap **GP** between the first substrate **101** and the second substrate **102** is defined by a thickness of the supporting structure **180**. In this embodiment, the supporting structure **180** is used to support the gap **GP** between the first substrate **101** and the second substrate **102**, so as to prevent the gap **GP** between the first substrate **101** and the second substrate **102** from collapsing.

In some embodiments, a height **180h** of the supporting structure **180** is greater than or equal to a sum of a height **h1** of the first microlenses **ML1** and a height **h2** of the second microlenses **ML2**, but the disclosure is not limited thereto. In other embodiments, the dummy first microlenses **ML1** are disposed below the supporting structure **180**, and the supporting structure **180** is located on the dummy first microlenses **ML1**. At this time, the height **180h** of the supporting structure **180** is greater than or equal to the height **h2** of the second microlenses **ML2**. In some embodiments, a thickness of the third collimating structure **CM3** is relatively thin and thus may be omitted.

The first microlenses **ML1** and the second microlenses **ML2** are respectively disposed on a first side **S1** and a

second side **S2** of the gap **GP**. In this embodiment, the first side **S1** of the gap **GP** is a side of the gap **GP** close to the first substrate **101**, and the second side **S2** is a side of the gap **GP** close to the second substrate **102**. In other words, the first microlenses **ML1** are disposed on the side of the gap **GP** close to the first substrate **101**, and the second microlenses **ML2** are disposed on the side of the gap **GP** close to the second substrate **102**. In other embodiments, positions of the first microlenses **ML1** and the second microlenses **ML2** may be interchanged with one another.

In this embodiment, the first photosensitive element **112** is overlapping with one of the first microlenses **ML1** and one of the second microlenses **ML2** in the direction **Z**. The second photosensitive element **114** is overlapping with one of the second microlenses **ML2** in the direction **Z**, and is not overlapping with the first microlenses **ML1**. The third photosensitive element **116** is overlapping with one of the first microlenses **ML1** in the direction **Z**, and is not overlapping with the second microlenses **ML2**.

In this embodiment, the first microlenses **ML1** and the second microlenses **ML2** are convex lenses, and a focus length of the convex lens may be calculated by Formula 1.

$$\frac{1}{f} = \frac{(n-1)}{nR} \quad \text{Formula 1}$$

In Formula 1, *f* is a focal length of the microlens; *n* is a refractive index, and *R* is a radius of curvature of the microlens.

In this embodiment, the light **L1**, the light **L2**, and the light **L3** each have different wavelengths. The longer the wavelength of the light, the less the refractive index *n* in the medium, and the easier it is to be focused at a deeper position. In this embodiment, since a radius of curvature **R1** of the first microlenses **ML1** is different from a radius of curvature **R2** of the second microlenses **ML2**, the light **L1** and the light **L2** are prevented from being focused at the position with different depths. Specifically, in this embodiment, the third wavelength λ_3 of the light **L3** is greater than the second wavelength λ_2 of the light **L2**. Therefore, when a material of the first microlenses **ML1** is the same as a material of the second microlenses **ML2**, a refractive index *n3* of the light **L3** in the first microlenses **ML1** is less than a refractive index *n2* of the light **L2** in the second microlenses **ML2**. The radius of curvature **R2** of the second microlenses **ML2** is greater than the radius of curvature **R1** of the first microlenses **ML1**, so as to improve an issue that the focus position of the light **L3** is deeper than the focus position of the light **L2** due to the refractive index *n3* being less than the refractive index *n2*. In this embodiment, the light **L3** passing through the first microlenses **ML1** is focused at a position of the third photosensitive element **116**, and the light **L2** passing through the second microlenses **ML2** is focused at a position of the second photosensitive element **114**.

In addition, when the first microlenses **ML1** and the second microlenses **ML2** are combined together, the focus length may be calculated by Formula 2.

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2} \quad \text{Formula 2}$$

In Formula 2, *f* is the focal length after the first microlenses **ML1** and the second microlenses **ML2** are combined

together; *f1* is the focal length of the first microlenses **ML1**; *f2* is the focal length of the second microlenses **ML2**, and *d* is a vertical spacing between the first microlenses **ML1** and the second microlenses **ML2**. In this embodiment, *d* is much less than *f1* and *f2*, or even *d* is equal to 0. Therefore, *d/f1f2* approaches 0. For example, *d* is less than 1 micrometer, and *f1* and *f2* are respectively 10 micrometers to 25 micrometers. Therefore, Formula 2 may be further omitted to be Formula 3.

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} \quad \text{Formula 3}$$

In this embodiment, the first wavelength λ_1 of the light **L1** is greater than the third wavelength λ_3 of the light **L3** and the second wavelength λ_2 of the light **L2**. Therefore, when the material of the first microlenses **ML1** is the same as the material of the second microlenses **ML2**, a refractive index *n1* of the light **L1** in the first microlenses **ML1** or the second microlenses **ML2** is less than the refractive index *n3* of the light **L3** in the first microlenses **ML1** and the refractive index *n2* of the light **L2** in the second microlenses **ML2**. The first microlenses **ML1** are overlapping with the second microlenses **ML2**, so as to improve an issue that the focus position of the light **L1** is too deep due to the first wavelength λ_1 being greater than the third wavelength λ_3 and the second wavelength λ_2 . In this embodiment, the light **L1** passing through the first microlenses **ML1** and the second microlenses **ML2** is focused at a position of the first photosensitive element **112**.

Based on the above, by disposing the first microlenses **ML1** and the second microlenses **ML2**, the light **L1**, the light **L2**, and the light **L3** may be respectively focused at the positions of the first photosensitive element **112**, the second photosensitive element **114**, and the third photosensitive element **116**, thereby enhancing the image sensing quality. In addition, all the first microlenses **ML1** are formed on the same layer by the same process, and all the second microlenses **ML2** are formed on the same layer by another same process. In other words, in this embodiment, the first microlenses **ML1** and the second microlenses **ML2** having different radii of curvature are respectively formed in different layers, which may have a lower manufacturing cost compared with forming the first microlenses **ML1** and the second microlenses **ML2** having different radii of curvature in the same layer.

In addition, although in FIG. 1, the photosensitive device **100** is configured to receive the light **L1**, the light **L2**, and the light **L3** having three different wavelengths, the disclosure is not limited thereto. In other embodiments, the photosensitive device is configured to receive at least two of the light **L1**, the light **L2**, and the light **L3**. In other words, in other embodiments, a method of disposing the microlenses includes at least two of the photosensitive elements overlapping with both the first microlenses **ML1** and the second microlenses **ML2**, the photosensitive elements only overlapping with the first microlenses **ML1**, and the photosensitive elements only overlapping with the second microlenses **ML2**. In other words, the disclosure does not limit the photosensitive device **100** to have the three features of the photosensitive elements overlapping with both the first microlenses **ML1** and the second microlenses **ML2**, the photosensitive elements only overlapping with the first

microlenses ML1, and the photosensitive elements only overlapping with the second microlenses ML2 at the same time.

FIG. 3 is a schematic cross-sectional view of a display device according to an embodiment of the disclosure. It is noted that some of the reference numerals and descriptions of the embodiment of FIG. 1 will apply to the embodiment of FIG. 3. The same or similar reference numerals will represent the same or similar elements, and the descriptions of the same technical contents will be omitted. Reference may be made to the above embodiment for the omitted descriptions, which will not be repeated in the following embodiments.

A difference between a display device 20 of FIG. 3 and the display device 10 of FIG. 1 is that in the display device 20, the radius of curvature R1 of the first microlenses ML1 is equal to the radius of curvature R2 of the second microlenses ML2, and the material of the first microlenses ML1 is different from the material of the second microlenses ML2.

In some embodiments, the material of the first microlenses ML1 is adjusted to increase a refractive index n_3' of the light L3 in the first microlenses ML1 (e.g., greater than the refractive index n_3 of the light L3 in the first microlenses ML1 in the display device 10). In other embodiments, the material of the second microlenses ML2 is adjusted to reduce a refractive index n_2' of the light L2 in the second microlenses ML2 (e.g., less than the refractive index n_2 of the light L2 in the second microlenses ML2 in the display device 10). In some embodiments, both the material of the first microlenses ML1 and the material of the second microlenses ML2 are adjusted.

In this embodiment, under the light having the same wavelength, the refractive index of the first microlenses ML1 is greater than the refractive index of the second microlenses ML2, and the refractive index of the light L3 having the longer wavelength in the first microlenses ML1 is approximately equal to the refractive index of the light L2 having the shorter wavelength in second microlenses ML2. Based on this, the issue that a focus depth of the light L1 is deeper than a focus depth of the light L2 due to the fact that the third wavelength λ_3 is greater than the second wavelength λ_2 may be improved.

In addition, although in FIG. 3, the photosensitive device 100 is configured to receive the light L1, the light L2, and the light L3 having three different wavelengths, the disclosure is not limited thereto. In other embodiments, the photosensitive device is configured to receive at least two of the light L1, the light L2, and the light L3. In other words, in other embodiments, a method of disposing the microlenses includes at least two of the photosensitive elements overlapping with both the first microlenses ML1 and the second microlenses ML2, the photosensitive elements only overlapping with the first microlenses ML1, and the photosensitive elements only overlapping with the second microlenses ML2. In other words, the disclosure does not limit the photosensitive device 100 to have the three features of the photosensitive elements overlapping with both the first microlenses ML1 and the second microlenses ML2, the photosensitive elements only overlapping with the first microlenses ML1, and the photosensitive elements only overlapping with the second microlenses ML2 at the same time.

FIG. 4 is a schematic cross-sectional view of a display device according to an embodiment of the disclosure. It is noted that some of the reference numerals and descriptions of the embodiment of FIG. 1 will apply to the embodiment of FIG. 4. The same or similar reference numerals will

represent the same or similar elements, and the descriptions of the same technical contents will be omitted. Reference may be made to the above embodiment for the omitted descriptions, which will not be repeated in the following embodiments.

A difference between a display device 30 of FIG. 4 and the display device 10 of FIG. 1 is that in the display device 30, the first side S1 of the gap GP where the first microlenses ML1 are disposed is the side of the gap GP close to the second substrate 102, and the second side S2 of the gap GP where the second microlenses ML2 are disposed is the side of the gap GP close to the first substrate 101.

In this embodiment, the first microlenses ML1 are disposed on the second substrate 102, and the second microlenses ML2 are disposed on the first substrate 101.

In this embodiment, the radius of curvature R1 of the first microlenses ML1 is less than the radius of curvature R2 of the second microlenses ML2. In this embodiment, the material of the first microlenses ML1 is the same as the material of the second microlenses ML2, but the disclosure is not limited thereto. In other embodiments, the material of the first microlenses ML1 is different from the material of the second microlenses ML2.

In addition, although in FIG. 4, the photosensitive device 100 is configured to receive the light L1, the light L2, and the light L3 having three different wavelengths, the disclosure is not limited thereto. In other embodiments, the photosensitive device is configured to receive at least two of the light L1, the light L2, and the light L3. In other words, in other embodiments, a method of disposing the microlenses includes at least two of the photosensitive elements overlapping with both the first microlenses ML1 and the second microlenses ML2, the photosensitive elements only overlapping with the first microlenses ML1, and the photosensitive elements only overlapping with the second microlenses ML2. In other words, the disclosure does not limit the photosensitive device 100 to have the three features of the photosensitive elements overlapping with both the first microlenses ML1 and the second microlenses ML2, the photosensitive elements only overlapping with the first microlenses ML1, and the photosensitive elements only overlapping with the second microlenses ML2 at the same time.

FIG. 5 is a schematic cross-sectional view of a display device according to an embodiment of the disclosure. It is noted that some of the reference numerals and descriptions of the embodiment of FIG. 1 will apply to the embodiment of FIG. 5. The same or similar reference numerals will represent the same or similar elements, and the descriptions of the same technical contents will be omitted. Reference may be made to the above embodiment for the omitted descriptions, which will not be repeated in the following embodiments.

A difference between a display device 40 of FIG. 5 and the display device 10 of FIG. 1 is that in the display device 40, the first photosensitive element 112 is overlapping with one of the first microlenses ML1 and one of the second microlenses ML2, and in a combination of the first photosensitive element 112 overlapping with the first microlenses ML1 and the second microlenses ML2, the vertical spacing d between the first microlenses ML1 and the second microlenses ML2 is equal to the focal length f_1 of the first microlenses ML1 plus the focal length f_2 of the second microlenses ML2.

In this embodiment, the first wavelength λ_1 of the light L1 is greater than the second wavelength λ_2 of the light L2. It may be known from Formula 2 that when the vertical spacing d is equal to the focal length f_1 plus the focal length

11

f2, the focal length f after the first microlenses ML1 and the second microlenses ML2 are combined together is nearly infinite. In other words, the combination of first microlenses ML1 and second microlenses ML2 may convert the light L1 into parallel light. In this embodiment, because the wavelength of the light L1 is too long, the microlenses exceed the process capability, and it is difficult to focus the light. However, through the combination of the first microlenses ML1 and the second microlenses ML2, the convergence of the light L1 may be improved, and the collimation effect may be obtained, thereby improving the imaging quality.

In this embodiment, the first microlenses ML1 are closer to the first photosensitive element 112 than the second microlenses ML2. The focal length f2 of the second microlenses ML2 is greater than the focal length f1 of the first microlenses ML1, and a magnification (M) of the combination of the first microlenses ML1 and the second microlenses ML2 is equal to $-f1/f2$, and $-f1/f2$ is less than -1 . In this embodiment, the radius of curvature of the first microlenses ML1 is the same as the radius of curvature of the second microlenses ML2, but the disclosure is not limited thereto. In other embodiments, the radius of curvature of the first microlenses ML1 is different from the radius of curvature of the second microlenses ML2. In this embodiment, the material of the first microlenses ML1 is the same as the material of the second microlenses ML2, but the disclosure is not limited thereto. In other embodiments, the material of the first microlenses ML1 is different from the material of the second microlenses ML2.

In some embodiments, the height 180h of the supporting structure 180 is approximately equal to a sum of the vertical spacing d, the height h1 of the first microlenses ML1, and the height h2 of the second microlenses ML2, but the disclosure is not limited thereto. In other embodiments, the dummy first microlenses ML1 are disposed below the supporting structure 180, and the supporting structure 180 is located on the dummy first microlenses ML1. At this time, the height 180h of the supporting structure 180 is approximately equal to a sum of the vertical spacing d and the height h2 of the second microlenses ML2.

In addition, although in FIG. 5, the second photosensitive element 114 is overlapping with the first microlenses ML1, but is not overlapping with the second microlenses ML2. However, the disclosure is not limited thereto. In other embodiments, the second photosensitive element 114 is overlapping with the second microlenses ML2, but is not overlapping with the first microlenses ML1.

Based on the above, in the photosensitive device of the disclosure, by disposing the first microlenses and the second microlenses, the issue of different focus depths caused by the inconsistent wavelengths of the light may be improved, thereby enhancing the image sensing quality.

FIG. 6 is a schematic cross-sectional view of a display device according to an embodiment of the disclosure. It is noted that some of the reference numerals and descriptions of the embodiment of FIG. 1 will apply to the embodiment of FIG. 6. The same or similar reference numerals will represent the same or similar elements, and the descriptions of the same technical contents will be omitted. Reference may be made to the above embodiment for the omitted descriptions, which will not be repeated in the following embodiments.

Referring to FIG. 6, in this embodiment, the first photosensitive element 112, the second photosensitive element 114, and the third photosensitive element 116 are each overlapping with one of the first microlenses ML1 and one of the second microlenses ML2.

12

In this embodiment, the first microlenses ML1 and the second microlenses ML2 respectively have a first radius of curvature R1 and a second radius of curvature R2, and the first radius of curvature R1 and the second radius of curvature R2 may optionally be the same, but the disclosure is not limited thereto.

For example, a display device 50 may further be provided with the cover plate 250 on a side of the display panel 200 away from the photosensitive device 100 (or a side of the second substrate 102 away from the first substrate 101), and the cover plate 250 has a cover plate surface 250s away from the second substrate 102 (or the display panel 200). The light shielding pattern layer LS further has a surface LS1s facing the second substrate 102. There is a distance D0 between the cover plate surface 250s of the cover plate 250 and the surface LS1s of the light shielding pattern layer LS, and there is a distance D1 between a surface 171s of the cladding layer 171 and the surface LS1s of the light shielding pattern layer LS. The voids LSa of the light shielding pattern layer LS and the second microlenses ML2 respectively have a length L1 and a length L2 along an arrangement direction (e.g., the direction X). The second microlenses ML2 further have the height h2 along a direction perpendicular to the arrangement direction.

In an exemplary embodiment, the second radius of curvature R2 of the second microlenses ML2 may satisfy the following relational Formula:

$$\frac{D_0 + D_1}{2} \geq R2 \geq \frac{4h2^2 + L_2^2}{8h2},$$

and the length L1 of the voids LSa may satisfy the following relational Formula:

$$L_2 \geq L_1 \geq \left(\frac{D_0}{D_0 + D_1} \right) \cdot L_2.$$

On the other hand, the first collimating structure CM1 further has a surface CM1s facing the gap GP. In this embodiment, the planar layer 166 is located on the surface CM1s of the first collimating structure CM1. The first photosensitive element 112 to the third photosensitive element 116 each have a light receiving surface 115rs facing the gap GP. There is a distance D5 between a surface 166s of the planar layer 166 and the surface CM1s of the first collimating structure CM1, and there is a distance D6 between the surface CM1s of the first collimating structure CM1 and the light receiving surfaces 115rs of the first photosensitive element 112 to the third photosensitive element 116. The first microlenses ML1 and the first voids CM1a respectively have a length L3 and a length L4 (i.e., an aperture) along the arrangement direction (e.g., the direction X). The first microlenses ML1 further have the height h1 along the direction perpendicular to the arrangement direction. The first photosensitive element 112 to the third photosensitive element 116 have a length L5 along the direction X. The length L5 here is, for example, defined by a length of an interface between the second electrode E2 of the first photosensitive element 112 to the third photosensitive element 116 and the photoelectric conversion layer PCL along the direction X.

13

In an exemplary embodiment, the first radius of curvature **R1** of the first microlenses **ML1** may satisfy the following relational Formula:

$$\frac{D_5 + D_6}{2} \geq R1 \geq \frac{4h1^2 + L_3^2}{8h1},$$

and the length L_4 of the first voids **CM1a** may satisfy the following relational Formula:

$$L_3 \geq L_4 \geq \left(\frac{D_5}{D_5 + D_6} \right) \cdot L_3 \geq L_5.$$

It should be noted that multiple spacers **180** are further disposed between the cladding layer **171** and the flat planar **166**, and the spacers **180**, the surface **171s** of the cladding layer **171**, and the surface **166s** of the planar layer **166** define the gap **GP** that may accommodate the microlenses. More specifically, the first microlenses **ML1** and the second microlenses **ML2** are spaced apart from one another in a normal direction (e.g., the direction **Z**) of the surface **171s** of the cladding layer **171** due to the gap **GP**. The gap **GP** here may be a space filled with air, a specific gas, or in a near-vacuum state.

In this embodiment, the first microlenses **ML1** and the second microlenses **ML2** that are correspondingly disposed are spaced apart along the direction **Z** by the vertical spacing **d**, and the vertical distance **d** satisfies the following relational Formula:

$$d \geq 2 \frac{R1 \times R2}{R1 + R2}.$$

When the first radius of curvature **R1** of the first microlenses **ML1**, the second radius of curvature **R2** of the second microlenses **ML2**, and the vertical spacing **d** between the first microlenses **ML1** and the second microlenses **ML2** are designed within the above range, multiple streams of fingerprint image light **FPI** is transmitted in the form of parallel light in the gap **GP**. Therefore, a variation of a gap thickness (e.g. the vertical spacing **d**) of the gap **GP** does not affect the signal quality of the fingerprint image.

For example, the fingerprint image light **FPI** incident on the photosensitive device **100** at an appropriate angle passes through the first voids **CM1a** of the first collimating structure **CM1** and is transmitted to the corresponding light receiving surfaces **115rs** of the first photosensitive element **112** to the third photosensitive element **116** after passing through the voids **LSa** of the light shielding pattern layer **LS** and being refracted by the second microlenses **ML2** and the first microlenses **ML1**. The light receiving surface **115rs** here is defined by, for example, a surface of the second electrode **E2** of the first photosensitive element **112** to the third photosensitive element **116** facing the second substrate **102**. In other embodiments, the light receiving surface **115rs** may also be defined by a surface of the photoelectric conversion layer **PCL** of the first photosensitive element **112** to the third photosensitive element **116** facing the second substrate **102**.

On the contrary, a fingerprint image light **uFPI-A** (or an unintended external ambient light) incident on the photosensitive device **100** at a greater incident angle is blocked and may not be transmitted to the corresponding first pho-

14

tosensitive element **112** to third photosensitive element **116** after passing through the voids **LSa** of the light shielding pattern layer **LS** and being refracted by the second microlenses **ML2**. That is to say, the first microlenses **ML1** and the second microlenses **ML2** respectively disposed on the two substrates and corresponding to one another may reduce the incident angle of the light transmitted to the first photosensitive element **112** to the third photosensitive element **116**. That is, a design of using the first microlenses **ML1** together with the second microlenses **ML2** may limit a light receiving range of the photosensitive device **100**, and effectively suppress background noise (i.e., a sensing signal generated by the unintended light), so as to increase a signal-to-noise ratio (**SNR**) of a fingerprint signal. In addition, part of the light shielding pattern layer and the spacer layer (e.g., planar layer) therebetween used in a general fingerprint sensing module may also be replaced, which is helpful for simplifying a manufacturing process of the photosensitive device **100**.

On the other hand, the first microlenses **ML1** and the second microlenses **ML2** in the disclosure are disposed between the first substrate **101** and the second substrate **102**. Therefore, the microlenses may be prevented from being damaged by unintended impact or scratches in the subsequent process, which helps to increase a production yield rate and a process margin of the fingerprint sensing module.

In order to allow the photosensitive device **100** to have an anti-counterfeiting function, some of the voids **LSa** of the light shielding pattern layer **LS** may selectively be provided with multiple color filter patterns, for example, the first filter element **191**, the second filter element **192**, and the third filter element **193**, but the disclosure is not limited thereto. In other embodiments, the fingerprint sensing module may not be provided with the color filter patterns.

It should be noted that since the photosensitive device **100** in the disclosure is not provided with the microlens on a side surface facing the display panel **200**, it is suitable to adopt a direct bonding process to connect the display panel **200** and the photosensitive device **100**. In this way, the phenomenon of multiple reflections of the light between the display panel **200** and the photosensitive device **100** may be reduced, thereby greatly improving a fingerprint sensing signal of the display device **50**. For example, the photosensitive device **100** and the display panel **200** are bonded together by the optical adhesive layer **220** distributed over the entire surface.

What is claimed is:

1. A photosensitive device, comprising:

- a first substrate;
- a second substrate opposite to the first substrate, wherein there is a gap between the first substrate and the second substrate;
- a supporting structure located in the gap between the first substrate and the second substrate;
- a plurality of first microlenses and a plurality of second microlenses respectively disposed on a first side and a second side of the gap, wherein a pitch of the plurality of first microlenses and a pitch of the plurality of second microlenses are different;
- a first photosensitive element overlapping with one of the plurality of first microlenses and one of the plurality of second microlenses;
- a second photosensitive element overlapping with another one of the plurality of second microlenses;
- a first filter element overlapping with the first photosensitive element;

15

- a second filter element overlapping with the second photosensitive element; and
- a collimating structure located between the first substrate and the plurality of first microlenses, wherein the one of the plurality of first microlenses and the one of the plurality of second microlenses are located between the first photosensitive element and the first filter element, and the another one of the plurality of second microlenses is located between the second photosensitive element and the second filter element.
2. The photosensitive device according to claim 1, wherein the second photosensitive element is not overlapping with the plurality of first microlenses.
3. The photosensitive device according to claim 2, further comprising:
- a third photosensitive element overlapping with another one of the plurality of first microlenses and not overlapping with the plurality of second microlenses.
4. The photosensitive device according to claim 3, wherein the first filter element is configured to enable a light having a first wavelength to pass through, the second filter element is configured to enable the light having a second wavelength to pass through, the photosensitive device further comprises:
- a third filter element overlapping with the third photosensitive element and configured to enable the light having a third wavelength to pass through, wherein the first wavelength is greater than the third wavelength, and the third wavelength is greater than the second wavelength.
5. The photosensitive device according to claim 1, wherein a radius of curvature of each of the plurality of second microlenses is greater than a radius of curvature of each of the plurality of first microlenses.
6. The photosensitive device according to claim 5, wherein a material of each of the plurality of first microlenses is the same as a material of each of the plurality of second microlenses.
7. The photosensitive device according to claim 1, wherein under a light having a same wavelength, a refractive index of each of the plurality of first microlenses is greater than a refractive index of each of the plurality of second microlenses.
8. The photosensitive device according to claim 7, wherein a radius of curvature of the plurality of each of first microlenses is equal to a radius of curvature of each of the plurality of second microlenses.
9. The photosensitive device according to claim 1, wherein a vertical spacing (d) between the plurality of first microlenses and the plurality of second microlenses is less than 1 micrometer.
10. The photosensitive device according to claim 1, wherein the plurality of first microlenses are closer to the first photosensitive element than the plurality of second microlenses, a vertical spacing (d) between the plurality of first microlenses and the plurality of second microlenses is equal to a focal length (f1) of the plurality of first microlenses plus a focal length (f2) of the plurality of second microlenses, and the focal length (f2) is greater than the focal length (f1).
11. The photosensitive device according to claim 1, wherein the second photosensitive element is overlapping with another one of the plurality of first microlenses.

16

12. A photosensitive device, comprising:
- a first substrate;
- a second substrate opposite to the first substrate, wherein there is a gap between the first substrate and the second substrate;
- a plurality of first microlenses and a plurality of second microlenses respectively disposed on a first side and a second side of the gap, wherein a material of each of the plurality of first microlenses is different from a material of each of the plurality of second microlenses, a pitch of the plurality of first microlenses and a pitch of the plurality of second microlenses are different, and/or a radius of curvature of each of the plurality of first microlenses is different from a radius of curvature of each of the plurality of second microlenses;
- a first photosensitive element overlapping with one of the plurality of first microlenses;
- a second photosensitive element overlapping with one of the plurality of second microlenses;
- a first filter element overlapping with the first photosensitive element; and
- a second filter element overlapping with the second photosensitive element, wherein the one of the plurality of first microlenses is located between the first photosensitive element and the first filter element, and the one of the plurality of second microlenses is located between the second photosensitive element and the second filter element.
13. The photosensitive device according to claim 12, wherein the first photosensitive element and the second photosensitive element are located on the first substrate, the first side of the gap is a side of the gap close to the first substrate, and the second side of the gap is a side of the gap close to the second substrate.
14. The photosensitive device according to claim 12, wherein the first photosensitive element and the second photosensitive element are located on the first substrate, the first side of the gap is close to the second substrate, and the second side of the gap is close to the first substrate.
15. The photosensitive device according to claim 12, wherein the first photosensitive element is overlapping with the one of the plurality of first microlenses and another one of the plurality of second microlenses, the plurality of first microlenses are closer to the first photosensitive element than the plurality of second microlenses, a vertical spacing (d) between the plurality of first microlenses and the plurality of second microlenses is equal to a focal length (f1) of the plurality of first microlenses plus a focal length (f2) of the plurality of second microlenses, and the focal length (f2) is greater than the focal length (f1).
16. The photosensitive device according to claim 1, wherein an orthographic projection of the second photosensitive element on the first substrate does not overlap an orthographic projection of all of the plurality of first microlenses on the first substrate.
17. The photosensitive device according to claim 12, wherein an orthographic projection of the second photosensitive element on the first substrate does not overlap an orthographic projection of all of the plurality of first microlenses on the first substrate.

* * * * *